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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bromine oxidises methanoic acid to carbon dioxide.

The equation for the reaction is

$$HCOOH(aq) + Br_2(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$$

Which of the following methods would **not** be suitable for measuring the progress of this reaction?

- ☐ **A** colorimetry
- ☐ **B** measuring electrical conductivity
- ☐ **C** quenching and titrating with acid
- ☐ **D** measuring the volume of gas

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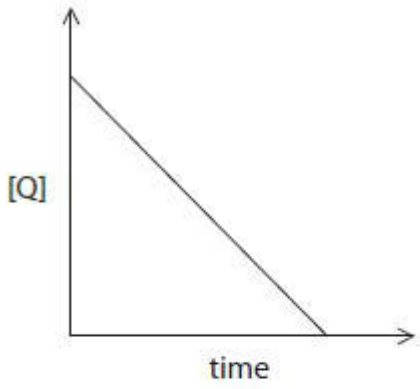
Q2.

A reactant Q is converted into two products. Which of these graphs shows that this reaction is first order with respect to Q?

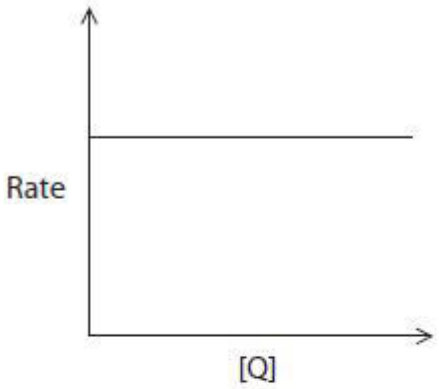
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Graph 1

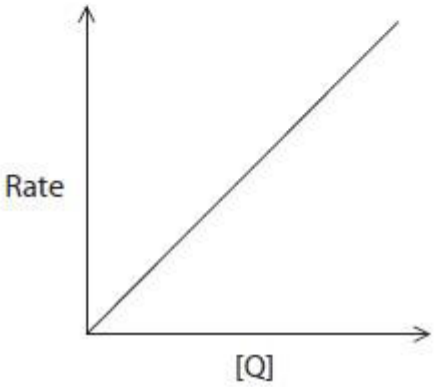


Graph 2

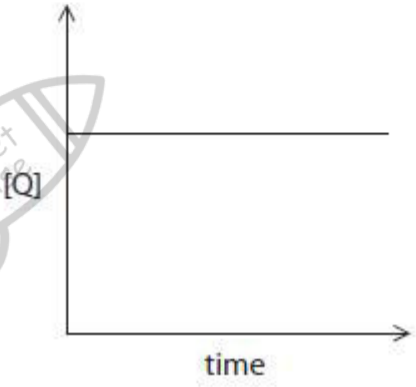


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Graph 3



Graph 4



☒ **A** Graph 1
☒ **B** Graph 2
☒ **C** Graph 3
☒ **D** Graph 4

(Total for question = 1 mark)

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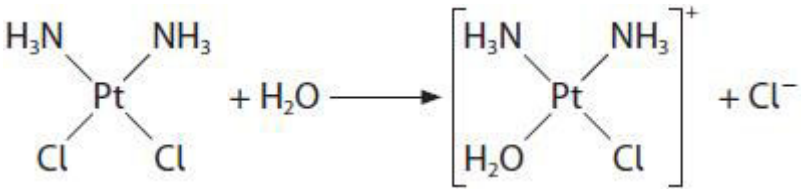
Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

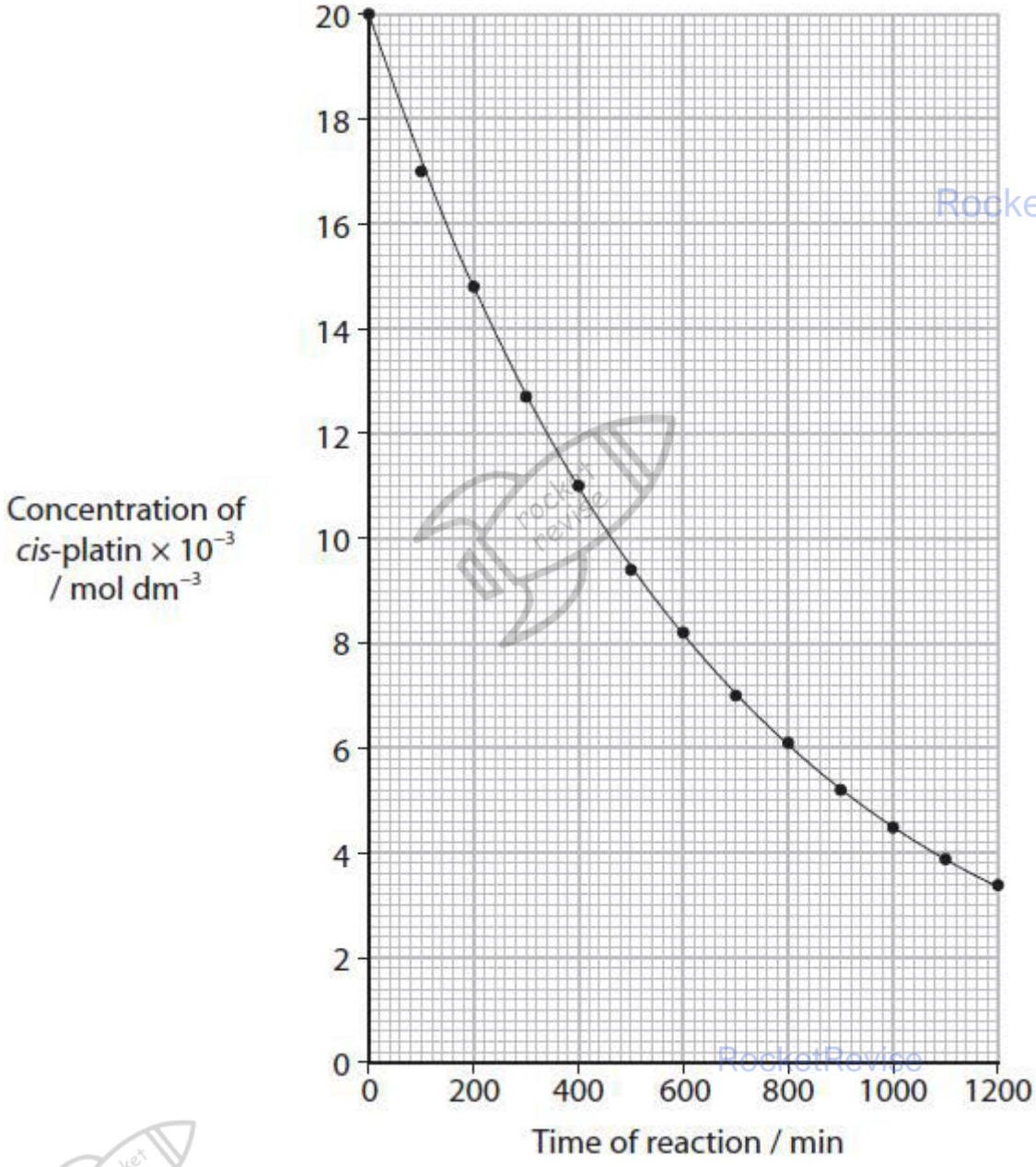
The inorganic anti-cancer drug *cis*-platin, $Pt(NH_3)_2Cl_2$, is hydrolysed by water to make it active. The reaction is

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The hydrolysis is first order overall. The half-life can be found from a graph of the concentration of *cis*-platin against time.



The half-life of the reaction is

- **A** 430 min
- **B** 460 min
- **C** 590 min
- **D** 600 min

(Total for question = 1 mark)

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Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The rate of the reaction between two compounds, **Y** and **Z**, was investigated.
The results are shown.

Experiment	Initial concentration of **Y** / mol dm^{-3}	Initial concentration of **Z** / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.64	0.24	8.00×10^{-3}
2	0.64	0.48	3.20×10^{-2}
3	0.32	0.48	3.20×10^{-2}

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What are the orders of reaction with respect to **Y** and **Z**?

	Order with respect to **Y**	Order with respect to **Z**
☐ **A**	0	1
☐ **B**	0	2
☐ **C**	1	1
☐ **D**	1	2

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(Total for question = 1 mark)

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Q5.

This question is about the kinetics of the reaction between nitrogen monoxide and oxygen.

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

The rate equation for the reaction is rate = $k[NO]^2[O_2]$

(a) Which statement is **not** correct?

(1)

- ☒ **A** the reaction is third order overall
- ☒ **B** the units of the rate of the reaction are mol dm^{-3} s^{-1}
- ☒ **C** the rate of the reaction increases when the pressure is increased
- ☒ **D** when the concentration of nitrogen monoxide doubles and the concentration of oxygen quadruples, the rate increases by a factor of 8

(b) What are the units of the rate constant, k, for this reaction?

(1)

- ☒ **A** dm^9 mol^{-3} s^{-1}
- ☒ **B** mol^3 dm^{-9} s^{-1}
- ☒ **C** mol^2 dm^{-6} s^{-1}
- ☒ **D** dm^6 mol^{-2} s^{-1}

(c) An experiment was carried out using 1.10×10^{-2} mol dm^{-3} of oxygen and some nitrogen monoxide.

The numerical value of the initial rate and rate constant were:

- initial rate = 3.20×10^{-3}
- rate constant, $k = 1.70 \times 10^3$

What was the initial concentration, in mol dm^{-3}, of the nitrogen monoxide used in the experiment?

(1)

- ☒ **A** 1.31×10^{-2}
- ☒ **B** 1.71×10^{-4}
- ☒ **C** 2.02×10^{-3}
- ☒ **D** 4.50×10^{-2}

(Total for question = 3 marks)

Q6.

The rate constant for the reaction to convert cyclopropane to propene was determined at five different temperatures.

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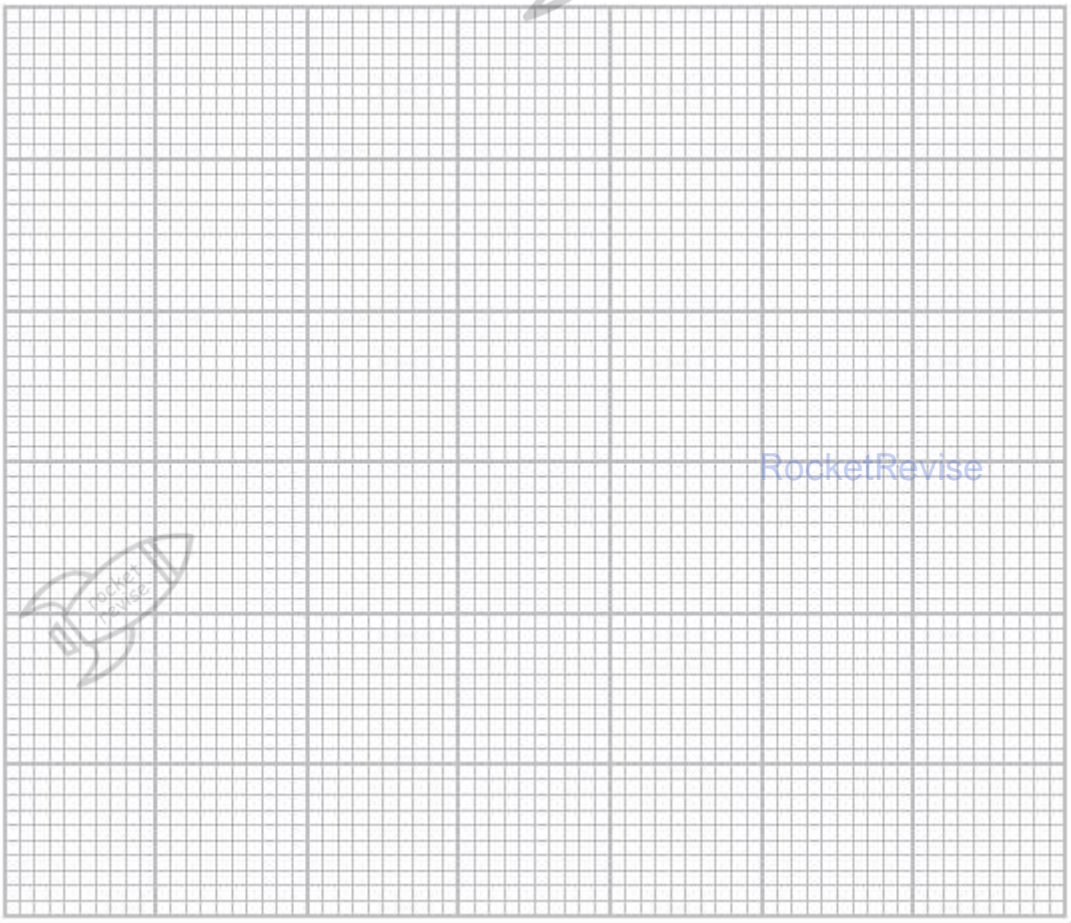


The results are shown in the table.

Temperature (T) /K	1 / Temperature ($1/T$) /K^{-1}	Rate constant (k) /s^{-1}	ln k
719	1.39×10^{-3}	2.49×10^{-5}	−10.60
746	1.34×10^{-3}	1.23×10^{-4}	−9.00
791	1.26×10^{-3}	1.66×10^{-3}	
840		1.83×10^{-2}	−4.00
889	1.12×10^{-3}	1.65×10^{-1}	−1.80

(a) (i) Complete the table. **(1)**

(ii) Plot a graph of ln k against $1 / T$. **(3)**



(iii) Use your graph to determine the activation energy, E_a, in kJ mol^{-1}.

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You should include the value and units of the gradient of the line.
The Arrhenius equation is

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant}$$

(3)

(b) The activation energy for many reactions is around +50 kJ mol^{-1}.

Given this information, comment on your value for E_a.

(2)

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(c) Explain the trend in the value of the rate constant k as the temperature increases.

(3)

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(Total for question = 12 marks)

Q7.

Which diagram shows the mechanism for the second order reaction between 1-bromoethane and potassium hydroxide in aqueous solution?

Diagram 1

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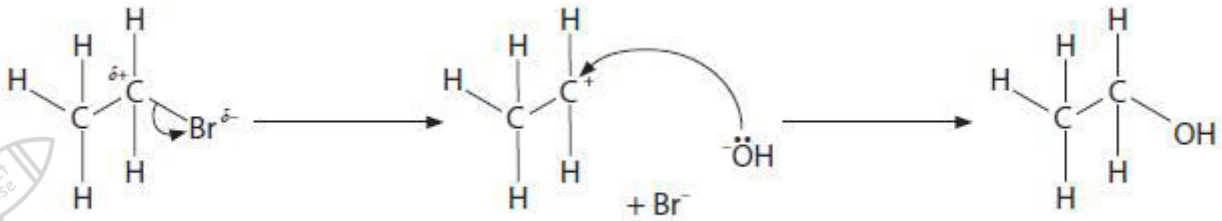


Diagram 2

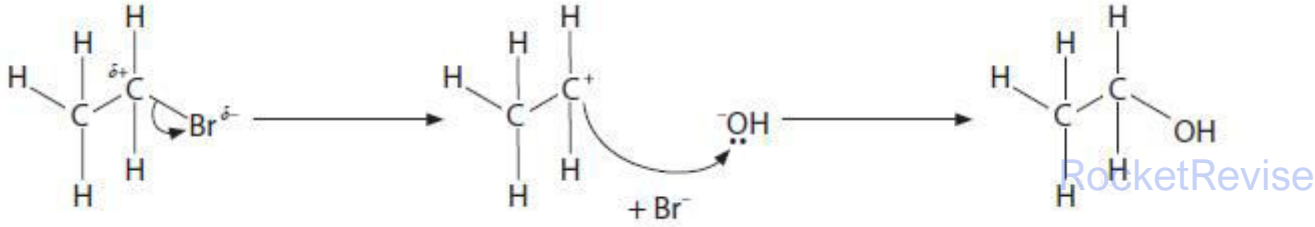


Diagram 3

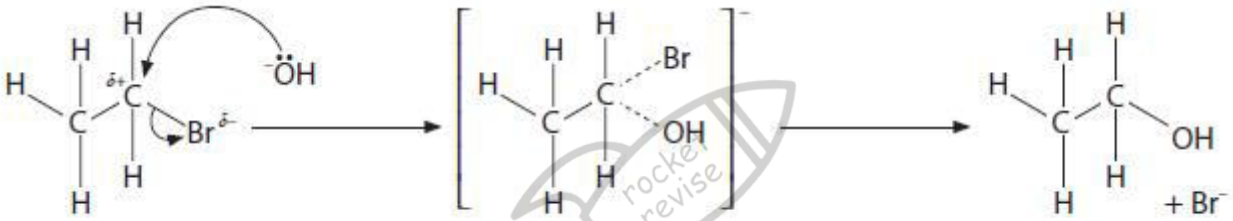
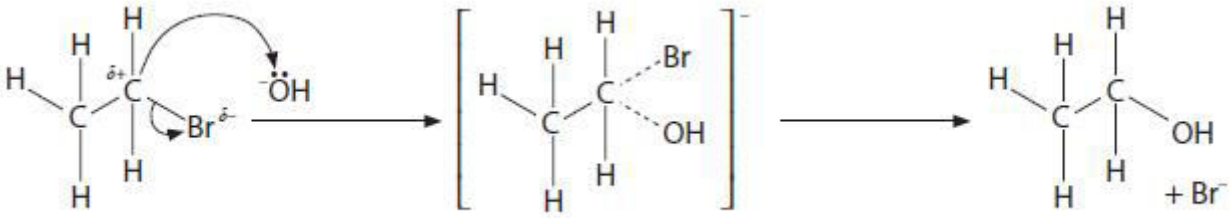


Diagram 4



- ☒ **A** Diagram 1
- ☒ **B** Diagram 2
- ☒ **C** Diagram 3
- ☒ **D** Diagram 4

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(Total for question = 1 mark)

Q8.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propanone reacts with iodine in acidic solution.

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$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$$

The rate equation for the formation of iodopropanone is found to be

$$\text{rate} = k[CH_3COCH_3(aq)][H^+(aq)]$$

(a) Which of the following is true?

(1)

- ☒ **A** the units for the rate constant are $dm^3\ mol^{-1}\ s^{-1}$
- ☒ **B** the reaction is a first order reaction overall
- ☒ **C** the units for the rate are $dm^3\ mol^{-1}\ s^{-1}$
- ☒ **D** doubling the concentrations of propanone and of iodine quadruples the rate

(b) Which of the following is **not** true?

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(1)

- ☒ **A** the reaction rate increases if the temperature is raised
- ☒ **B** the rate constant increases if the temperature is raised
- ☒ **C** the addition of a small amount of sodium hydroxide decreases the reaction rate
- ☒ **D** the rate is unchanged when the hydrogen ion concentration is doubled

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(Total for question = 2 marks)

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Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation shows the hydrolysis of a bromoalkane.

$$RBr + OH^- \rightarrow ROH + Br^-$$

The rate equation is rate = k[RBr]

RBr is most likely to be

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☒ **A** bromomethane
☒ **B** 2-bromopropane
☒ **C** 1-bromo-2-methylpropane
☒ **D** 2-bromo-2-methylpropane

(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The Arrhenius equation can be shown as

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant}$$

A graph is plotted of ln k against $1/T$ for a reaction.
The activation energy, E_a, of this reaction equals

☒ **A** – gradient ÷ R

☒ **B** + gradient ÷ R

☒ **C** – gradient × R

☒ **D** + gradient × R

(Total for question = 1 mark)

Q11.

The decomposition of benzenediazonium chloride is a first order reaction.

$$C_6H_5N_2Cl + H_2O \rightarrow C_6H_5OH + N_2 + HCl$$

The activation energy of this reaction was determined by measuring the rate constant at various temperatures.

(a) In an experiment at 333 K, the concentration of a sample of benzenediazonium chloride was measured at various times during its decomposition. The results of this experiment are shown.

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Time / s	$[C_6H_5N_2Cl]$ / mol dm^{-3}
0.0	0.500
40.0	0.410
100	0.285
200	0.165
280	0.100
350	0.070
400	0.050

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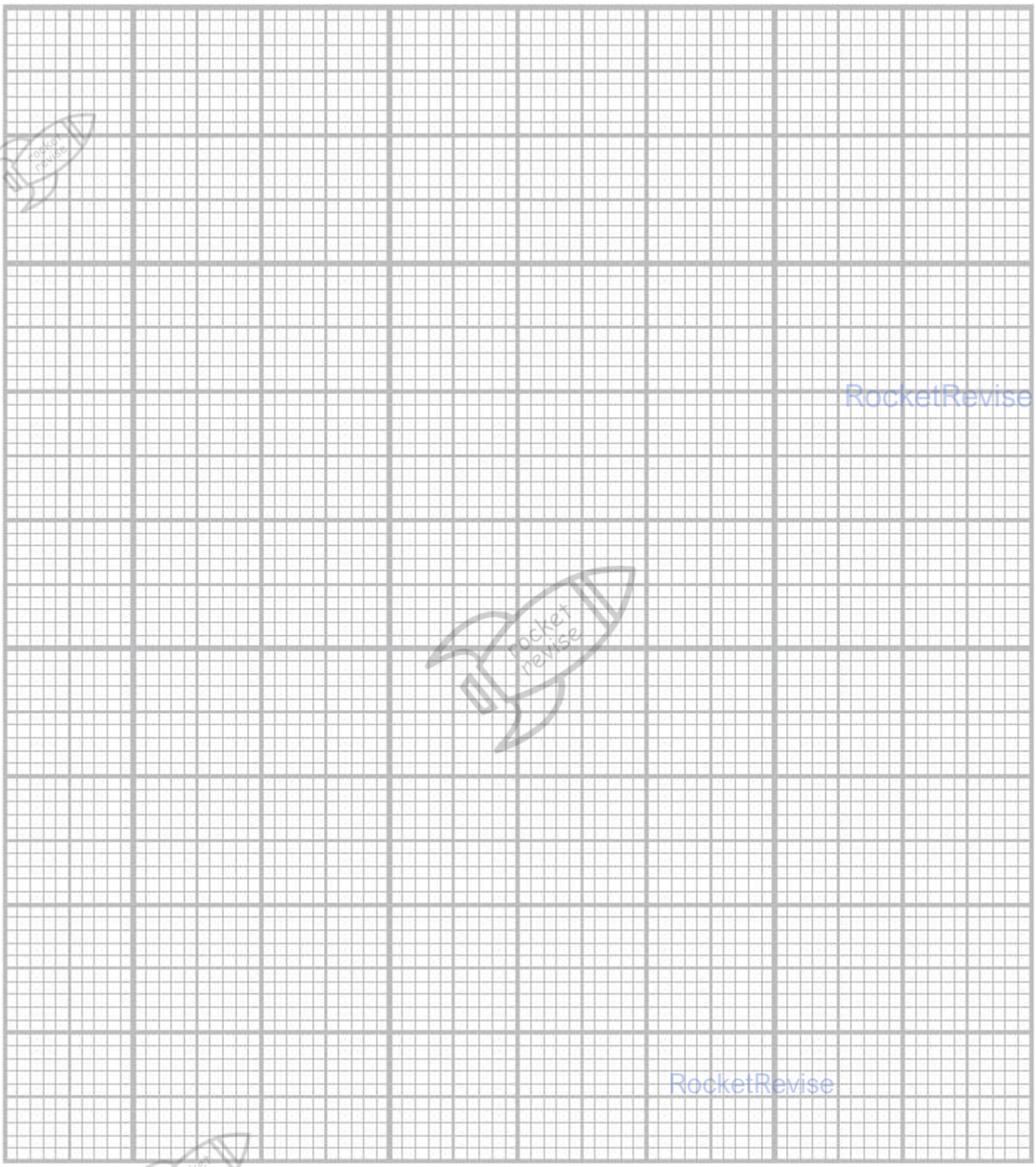
(i) Plot a graph of concentration of benzenediazonium chloride against time.

(3)

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(ii) Determine a value for the half-life, $t^1/_2$, of this reaction.
You **must** show your working on the graph.

(1)

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(iii) Calculate the rate constant, k, for the reaction at 333 K.
Include units in your answer.
Use the expression $\ln 2 = kt^1/_2$

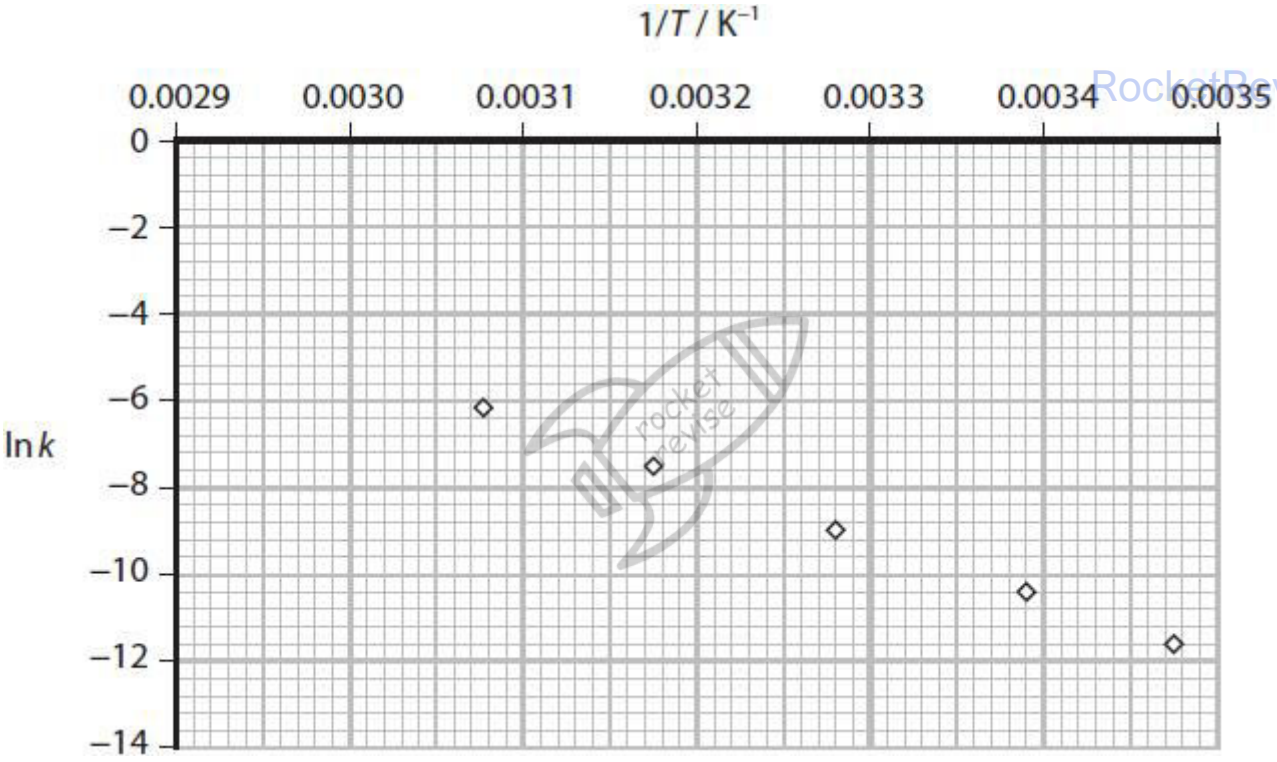
(2)

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(b) The experiment described in (a) was repeated for five temperatures and the data used to plot a graph for the Arrhenius equation in the form

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant}$$



(i) Use the rate constant that you have calculated in (a)(iii) to obtain data for a point on the graph for 333 K.

(2)

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(ii) Plot your data from (b)(i) on the graph.

(1)

(iii) Determine the gradient of the graph by drawing a best-fit line.
Include a sign and units in your answer.

(3)

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(iv) Use the gradient determined in (b)(iii) to calculate the activation energy for the decomposition of benzenediazonium chloride.

Include a sign and units with your answer.

(3)

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(Total for question = 15 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The halogenoalkane 2-bromo-2-methylbutane was hydrolysed with sodium hydroxide solution, NaOH(aq). Which suggestion about the mechanism of this reaction is correct?

	Type of mechanism	Number of steps in mechanism
☐ **A**	S_N2	one
☐ **B**	S_N2	two
☐ **C**	S_N1	one
☐ **D**	S_N1	two

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(Total for question = 1 mark)

Q13.

* The hydrolysis of halogenoalkanes by alkali is a nucleophilic substitution reaction.

$$RX + OH^- \rightarrow ROH + X^-$$

The mechanism of this reaction for primary halogenoalkanes is different from the mechanism for tertiary halogenoalkanes.

Describe how knowledge of the rate equations for the hydrolysis of halogenoalkanes provides evidence for the mechanisms of these reactions.

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Curly arrow mechanisms are **not** required.

(6)

(Total for question = 6 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Propanone reacts with iodine in the presence of a catalyst of dilute hydrochloric acid. The reaction occurs in aqueous solution.

$$CH_3COCH_3 + I_2 \rightarrow CH_3COCH_2I + HI$$

The rate equation for this reaction is

$$\text{rate} = k[CH_3COCH_3][H^+]$$

Which is a possible mechanism for the reaction?

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☒ **A**

$CH_3COCH_3 + H^+ \rightleftharpoons CH_3C(\overset{+}{O}H)CH_3$ fast

$CH_3C(\overset{+}{O}H)CH_3 \rightarrow CH_3C(OH)=CH_2 + H^+$ slow

$CH_3C(OH)=CH_2 + I_2 \rightarrow CH_3COCH_2I + HI$ fast

☒ **B**

$CH_3COCH_3 + H^+ \rightleftharpoons CH_3C(\overset{+}{O}H)CH_3$ fast

$CH_3C(\overset{+}{O}H)CH_3 \rightarrow CH_3C(OH)=CH_2 + H^+$ fast

$CH_3C(OH)=CH_2 + I_2 \rightarrow CH_3COCH_2I + HI$ slow

☒ **C**

$CH_3COCH_3 \rightarrow CH_3COCH_2^- + H^+$ slow

$I_2 \rightarrow I^+ + I^-$ slow

$CH_3COCH_2^- + I^+ \rightarrow CH_3COCH_2I$ fast

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☒ **D**

$I_2 \rightarrow I^+ + I^-$ slow

$CH_3COCH_3 + I^- \rightarrow CH_3COCH_2^- + HI$ slow

$CH_3COCH_2^- + I^+ \rightarrow CH_3COCH_2I$ fast

(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The rate equation for a reaction is

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$$\text{rate} = k[A]^2[B]^0$$

The initial rate of reaction is 9.0×10^{-5} mol dm^{-3} s^{-1} when [A] = 0.30 mol dm^{-3} and [B] = 0.20 mol dm^{-3}.

What is the value of the rate constant in dm^3 mol^{-1} s^{-1}?

☒ **A** 8.1×10^{-6}

☒ **B** 3.0×10^{-4}

☒ **C** 1.0×10^{-3}

☒ **D** 5.0×10^{-3}

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(Total for question = 1 mark)

Q16.

A student investigated the kinetics of the reaction between bromate(V) ions and bromide ions in acidic conditions.

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

(a) In the first experiment, the student measured the initial rate of the reaction at five different concentrations of bromate(V) ions, BrO_3^-. In each case, the initial concentrations of bromide ions and hydrogen ions were constant and in large excess. The results obtained are shown.

Initial concentration of bromate(V) ions / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
0.030	4.17×10^{-7}
0.060	8.34×10^{-7}
0.090	1.25×10^{-6}
0.120	1.67×10^{-6}
0.150	2.09×10^{-6}

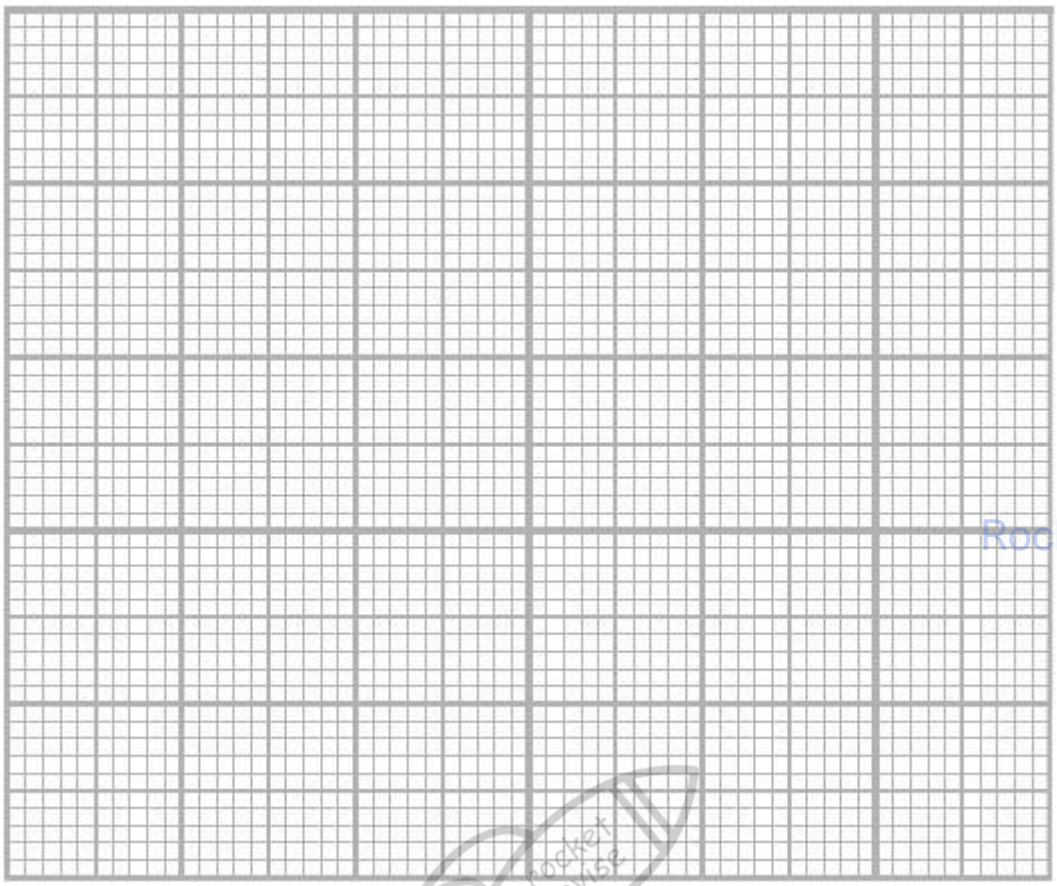
(i) Use the results to plot a suitable graph that can be used to show that the reaction is first order with respect to bromate(V) ions.

(3)

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(ii) State how your graph shows that the reaction is first order with respect to bromate(V) ions.

(1)

.....

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(b) In the second experiment, the student determined the initial rates of the same reaction starting with different concentrations of the reactants.

Run	$[BrO_3^-]$ / mol dm^{-3}	$[Br^-]$ / mol dm^{-3}	$[H^+]$ / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
1	0.062	0.21	0.40	1.52×10^{-5}
2	0.31	0.21	0.20	1.90×10^{-5}
3	0.062	0.63	0.40	4.56×10^{-5}

(i) Use these results and your answer to (a) to deduce the orders with respect to Br^- ions and H^+ ions.

(2)

Br^- ions

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H^+ ions

.....

(ii) Write the rate equation for the reaction.

(1)

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(iii) Use the results for Run 1 and your rate equation from (b)(ii) to calculate the value for the rate constant, k. Include units in your answer.

(3)

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(c) The presence of bromate(V) ions in drinking water is harmful to humans. Bromate(V) ions can be converted to less harmful bromide ions by passing the water through palladium with a reducing agent.

Describe how a heterogeneous catalyst, such as palladium, increases the rate of a reaction.

(3)

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(Total for question = 13 marks)

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Q17.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Nitrogen monoxide and hydrogen react together to form nitrogen and water.

$$2NO + 2H_2 \rightarrow N_2 + 2H_2O$$

The steps in the mechanism of the reaction are

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Step 1	$2NO \rightleftharpoons N_2O_2$	fast
Step 2	$N_2O_2 + H_2 \rightarrow N_2O + H_2O$	slow
Step 3	$N_2O + H_2 \rightarrow N_2 + H_2O$	fast

Which statement about the reaction is correct?

☐ **A** Step 3 is the rate determining step and the overall order is 2

☐ **B** Step 3 is the rate determining step and the overall order is 4

☐ **C** Step 2 is the rate determining step and the overall order is 2

☐ **D** Step 2 is the rate determining step and the overall order is 3

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(Total for question = 1 mark)

Q18.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The decomposition of ammonia is catalysed by tungsten metal.

$$2NH_3(g) \xrightarrow{\text{tungsten}} N_2(g) + 3H_2(g)$$

This reaction has zero order kinetics.

(a) What is the rate equation for this reaction?

(1)

☐ **A** rate = k

☐ **B** rate = $k[NH_3]$

☐ **C** rate = $k[NH_3]^2$

☐ **D** rate = $k[N_2][H_2]^3$

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(b) What are the units of the rate constant, k, for this zero order reaction?

(1)

☐ **A** no units

☐ **B** s^{-1}

☐ **C** mol dm^{-3} s^{-1}

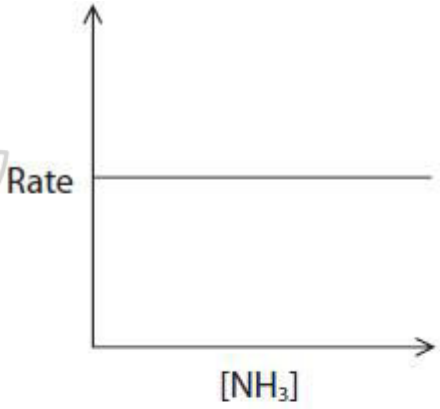
☐ **D** dm^3 mol^{-1} s^{-1}

(c) Which of these graphs represents this zero order reaction?

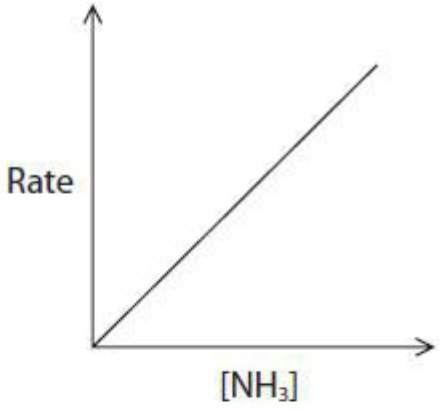
(1)

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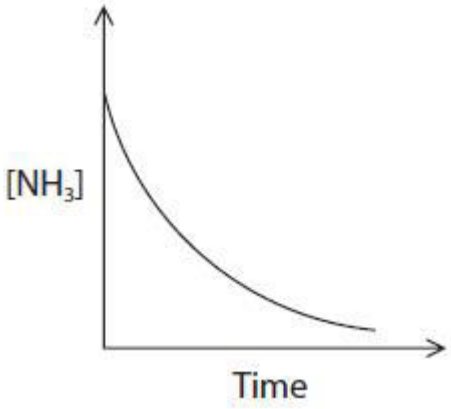


☒ **A**

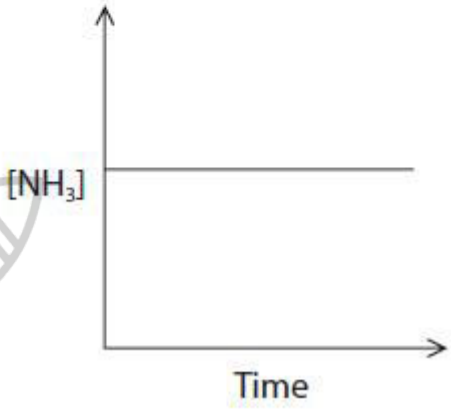


☒ **B**

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☒ **C**



☒ **D**

(Total for question = 3 marks)

Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which **continuous** monitoring method could be used to obtain rate data for the reaction shown?

$$C_4H_9Cl(l) + H_2O(l) \rightarrow C_4H_9OH(aq) + H^+(aq) + Cl^-(aq)$$

- ☒ **A** electrical conductivity
- ☒ **B** colorimetry
- ☒ **C** mass change
- ☒ **D** titration with alkali

(Total for question = 1 mark)

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Q20.

Azomethane, $CH_3N_2CH_3$, decomposes to form nitrogen and ethane.

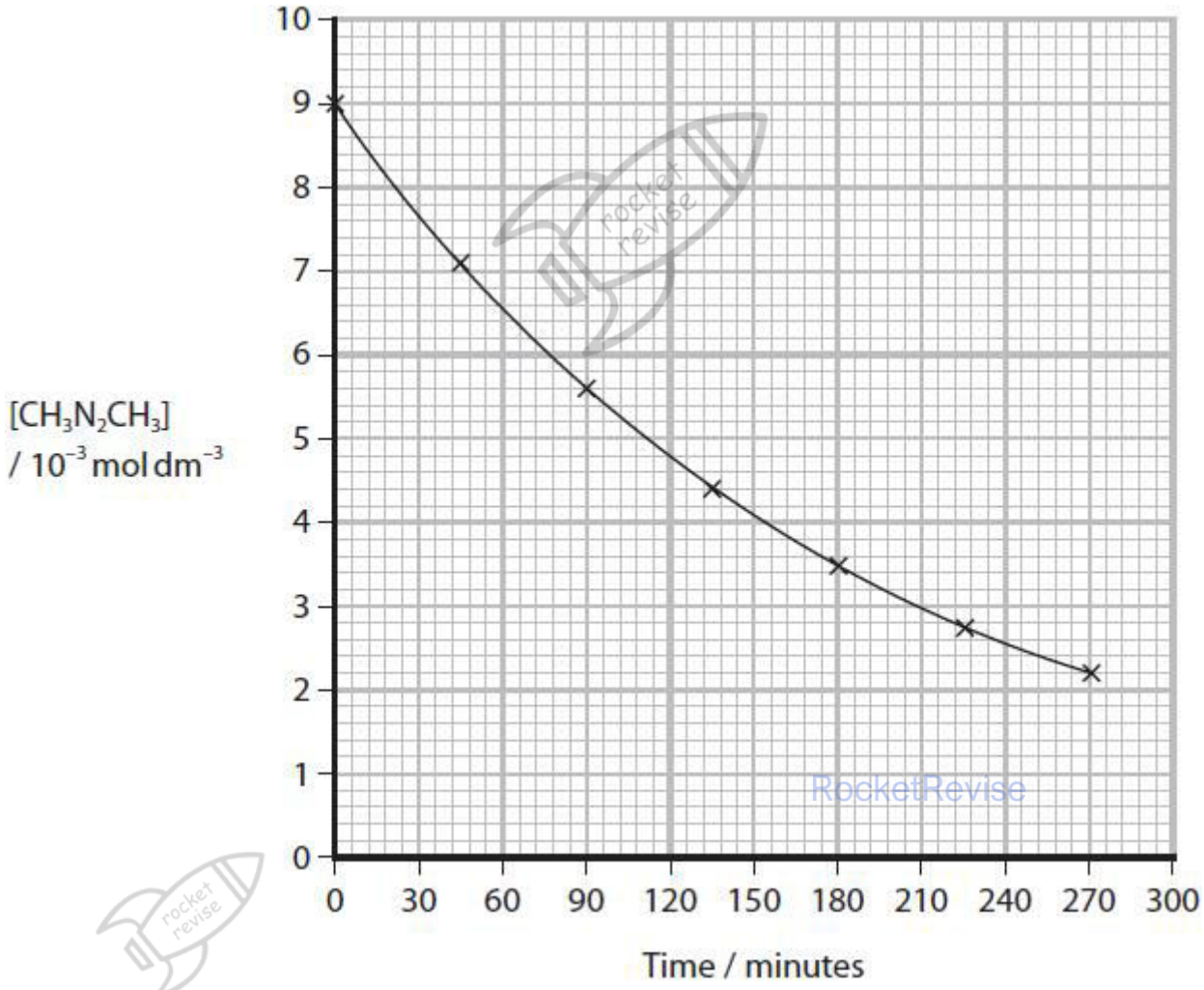
$$CH_3N_2CH_3(g) \rightarrow N_2(g) + C_2H_6(g)$$

(a) Write the rate equation for this first order reaction.

(1)

(b) The concentration of azomethane, $[CH_3N_2CH_3]$, was measured at various times during its decomposition at 573 K.

A graph of the data obtained is shown.



(i) Determine values for the half-life, $t_{1/2}$, of the decomposition of azomethane from the graph, stating whether or not these data confirm that the reaction is first order.

You must show your working on the graph.

(2)

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(ii) Calculate the rate constant, k, **in s^{-1}**, for the decomposition of azomethane at 573 K, using your

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answer to (b)(i) and the expression shown.

$$t_{1/2} = \frac{\ln 2}{k}$$

(2)

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(c) The rate constant for the decomposition of azomethane is 1.1×10^{-6} s^{-1} at 523 K and 3.5×10^{-3} s^{-1} at 623 K.

(i) Explain why the rate constant for this reaction is higher at 623 K than at 523 K.

(2)

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(ii) The rate constant changes with temperature according to the expression shown.

$$\log_{10}\left(\frac{k_1}{k_2}\right) = -\frac{E_a}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

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Calculate the activation energy, E_a, for the decomposition of azomethane using this expression and the values for k at 523 K and 623 K.
Include units and give your answer to **two** significant figures.
[R = 8.31 J mol^{-1} K^{-1}]

(3)

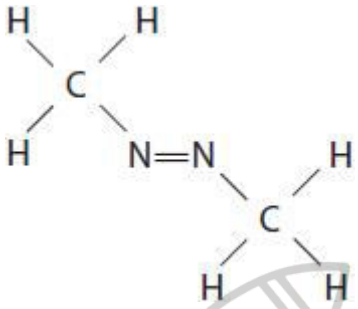
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(d) The activation energy for the decomposition of azomethane is high because the reaction involves covalent bond fission.

Identify on the structure of azomethane the covalent bond (or bonds) that break in this decomposition.

(1)



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(Total for question = 11 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A series of experiments was carried out to determine the rate constant, *k*, of the reaction between bromate(V) ions and bromide ions at different temperatures, *T*.

A graph of ln *k* was plotted against $\frac{1}{T}$. The gradient of the line was −4670 K.

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The Arrhenius equation may be written as

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant} \qquad R = 8.31\ \text{J mol}^{-1}\ \text{K}^{-1}$$

What is the activation energy of this reaction?

(1)

☒ **A** +38.8 kJ mol^{-1}
☒ **B** +38.8 J mol^{-1} K^{-1}
☒ **C** +562 J mol^{-1}
☒ **D** +562 J mol^{-1} K^{-1}

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(Total for question = 1 mark)

Q22.

This question is about the hydrolysis of 2-bromobutane with aqueous sodium hydroxide at 298 K.

The equation for the reaction is:

$$CH_3CH_2CHBrCH_3 + OH^- \rightarrow CH_3CH_2CHOHCH_3 + Br^-$$

(a) The data shown were obtained in an experiment to investigate the kinetics of this reaction.

Experiment number	Initial concentration / mol dm^{-3}		Initial rate / mol dm^{-3} s^{-1}
	[$CH_3CH_2CHBrCH_3$]	[OH^-]	
1	0.100	0.100	1.01×10^{-3}
2	0.200	0.100	2.02×10^{-3}
3	0.100	0.200	1.01×10^{-3}

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(i) Deduce the order of reaction with respect to 2-bromobutane and with respect to hydroxide ions. Justify your answers.

(2)

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(ii) Write the rate equation for the reaction using your answer to (a)(i).

(1)

(iii) Calculate the rate constant for Experiment 1.
Include units in your answer.

(2)

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(b) Draw the mechanism for this reaction that is consistent with your rate equation.

Include curly arrows, and any relevant dipoles and lone pairs of electrons.

(3)

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(Total for question = 8 marks)

Q23.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

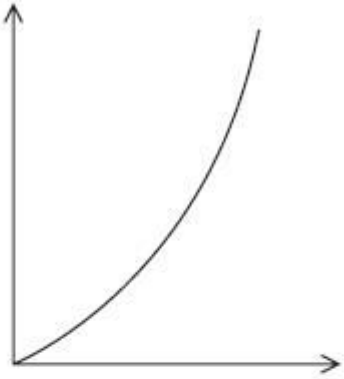
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Four graphs are shown.

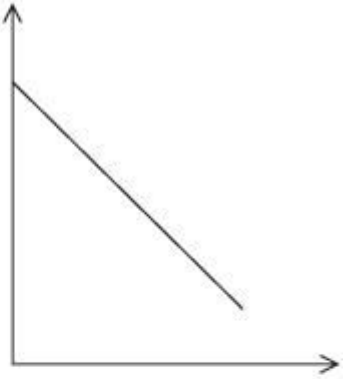
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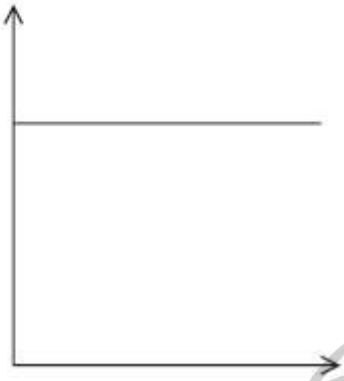
Graph 1



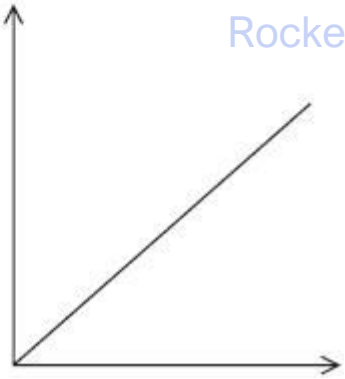
Graph 2



Graph 3



Graph 4



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(a) Which graph shows the rate of reaction on the vertical axis plotted against concentration of the reactant on the horizontal axis for a **zero** order reaction?

(1)

- ☒ **A** Graph 1
- ☒ **B** Graph 2
- ☒ **C** Graph 3
- ☒ **D** Graph 4

(b) Which graph shows the rate of reaction on the vertical axis plotted against concentration of the reactant on the horizontal axis for a **first** order reaction?

(1)

- ☒ **A** Graph 1
- ☒ **B** Graph 2
- ☒ **C** Graph 3
- ☒ **D** Graph 4

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(Total for question = 2 marks)

Q24.

Solid vanadium(V) oxide catalyses the equilibrium shown.

$2SO_2(g) + O_2(g) \leftrightarrow 2SO3(g)$ $\Delta H = -196$ kJ mol^{-1}

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Which statement is correct?

- ☒ **A** vanadium(V) oxide is a homogeneous catalyst
- ☒ **B** decreasing the pressure would increase the equilibrium yield of sulfur trioxide
- ☒ **C** increasing the surface area of the catalyst would increase the equilibrium yield of sulfur trioxide
- ☒ **D** decreasing the temperature would increase the equilibrium yield of sulfur trioxide

(Total for question = 1 mark)

Q25.

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Bromate(V) ions, BrO_3^-, oxidise bromide ions, Br^-, in dilute acid.

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

Experiments to determine initial reaction rates were carried out using different initial concentrations of the three reactants.

The results are shown in the table.

Experiment number	$[BrO_3^-(aq)]$ / mol dm^{-3}	$[Br^-(aq)]$ / mol dm^{-3}	$[H^+(aq)]$ / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
1	0.10	0.25	0.30	3.36×10^{-5}
2	0.10	0.25	0.60	1.34×10^{-4}
3	0.15	0.50	0.30	1.01×10^{-4}
4	0.15	0.25	0.60	2.01×10^{-4}

(a) The reaction is first order with respect to bromate(V) ions.

(i) Deduce the rate equation for the reaction.
Justify your answer using the data.

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(4)

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(ii) Use the data from Experiment 4 and your answer to (a)(i) to calculate the rate constant for the reaction. Include units in your answer.

(3)

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(b) Give **one** possible reason why the rate equation shows that the reaction cannot proceed in one step.

(1)

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(Total for question = 8 marks)

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Q26.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Nitrogen monoxide is reduced by hydrogen.

$$2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$$

The rate equation for the reaction is

$$\text{rate} = k\,[NO]^2\,[H^2]$$

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(a) The concentrations of **both** reactants are tripled.

By what factor does the rate of reaction increase?

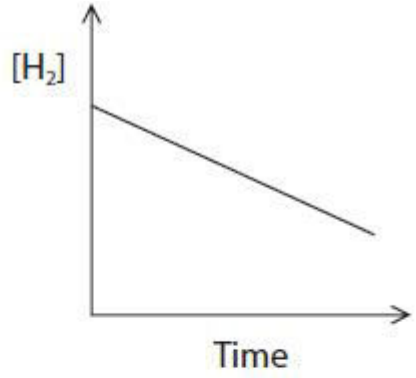
(1)

- ☒ **A** 3
- ☒ **B** 9
- ☒ **C** 18
- ☒ **D** 27

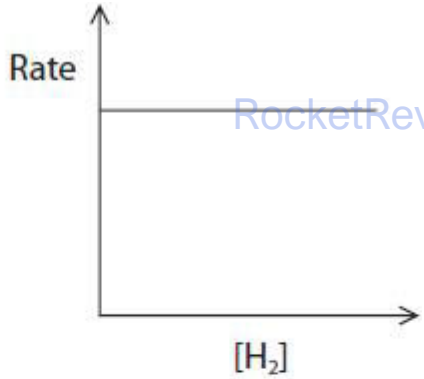
(b) Which graph is correct for this reaction?

(1)

☒ **A**

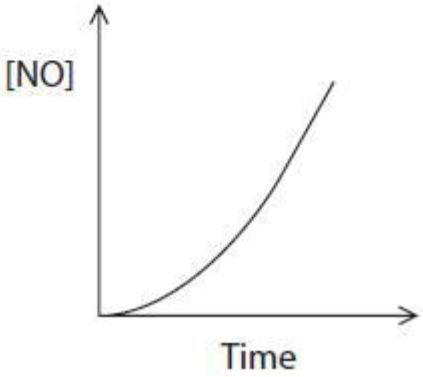


☒ **B**

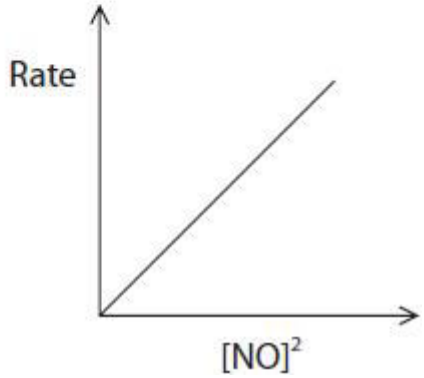


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☒ **C**



☒ **D**



(c) The reaction mechanism occurs in three steps.

Step **1** $NO + NO \leftrightarrow N_2O_2$
Step **2** $N_2O_2 + H_2 \rightarrow N_2O + H_2O$
Step **3** $N_2O + H_2 \rightarrow N_2 + H_2O$

Which is the rate-determining step in the mechanism?

(1)

- ☒ **A** Step **1**
- ☒ **B** Step **2**
- ☒ **C** Step **3**
- ☒ **D** cannot be deduced from the information provided

(Total for question = 3 marks)

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Q27.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two ways of writing an equation for the reaction between nitrogen monoxide and oxygen are shown.

$$2NO + O_2 \rightarrow 2NO_2 \qquad \text{Equation 1}$$

$$NO + \tfrac{1}{2}O_2 \rightarrow NO_2 \qquad \text{Equation 2}$$

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(a) Which **pair** of rate equations could be correct for this reaction?

(1)

		Equation 1	Equation 2
☐	**A**	$rate = k[NO]^2[O_2]$	$rate = k[NO]^2[O_2]$
☐	**B**	$rate = k[NO]^2[O_2]$	$rate = k[NO][O_2]^{\frac{1}{2}}$
☐	**C**	$rate = k[NO_2]^2$	$rate = k[NO_2]^2$
☐	**D**	$rate = k[NO_2]^2$	$rate = k[NO_2]$

(b) Which two methods can be used for **continuous** monitoring of the progress of this reaction?

(1)

- ☐ **A** colorimetry and titration
- ☐ **B** colorimetry and volume change
- ☐ **C** mass change and volume change
- ☐ **D** mass change and titration

(Total for question = 2 marks)

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Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The decomposition of hydrogen peroxide is catalysed by iodide ions.

$$2H_2O_2(aq) \xrightarrow{I^-} 2H_2O(l) + O_2(g)$$

The mechanism for this reaction is shown.

Step 1: $H_2O_2(aq) + I^-(aq) \rightleftharpoons IO^-(aq) + H_2O(l)$ fast

Step 2: $H_2O_2(aq) + IO^-(aq) \rightarrow I^-(aq) + H_2O(l) + O_2(g)$ slow

What is the rate equation for this reaction?

- ☒ **A** rate = $k[H_2O_2]^2[I^-]$
- ☒ **B** rate = $k[H_2O_2][I^-]$
- ☒ **C** rate = $k[H_2O_2]^2[I^{â€“}][IO^-]$
- ☒ **D** rate = $k[H_2O_2][IO^-]$

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(Total for question = 1 mark)

Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The half-life of a first order chemical reaction is

- ☒ **A** half the time taken for the reaction to be complete
- ☒ **B** the time taken for the value of the rate constant to halve
- ☒ **C** the time taken for the concentration of a reactant to halve
- ☒ **D** the time taken for the concentration of a product to double

(Total for question = 1 mark)

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Q30.

The reaction between hydrogen peroxide and iodide ions in acid conditions is known as the Harcourt–Esson reaction after the scientists who first studied its kinetics.

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow 2H_2O(l) + I_2(aq)$$

(a) A student carried out experiments to determine the rate equation for this reaction at 293 K.

(i) The rate of this reaction may be obtained by adding a fixed volume of sodium thiosulfate solution and a few drops of starch solution to the reaction mixture.

Explain how this method gives the rate of reaction.

(3)

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(ii) The student's results are shown.

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Run	$[H_2O_2(aq)]$ / mol dm^{-3}	$[I^-(aq)]$ / mol dm^{-3}	$[H^+(aq)]$ / mol dm^{-3}	Rate / mol dm^{-3} s^{-1}
1	0.0210	0.0198	0.00105	0.00181
2	0.0105	0.0400	0.00105	0.00181
3	0.0105	0.0797	0.00099	0.00364
4	0.0210	0.0801	0.00201	0.00730

Deduce the rate equation for this reaction.

(3)

(iii) Calculate the rate constant for this reaction at 293 K, using the results from Run 1. Include units with your answer.

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(3)

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(b) The student carried out a second series of experiments with this reaction at 313 K.

The rate of reaction was found to be 4.45 times faster at 313 K than at 293 K.
Calculate the activation energy, E_a, for this reaction, using the Arrhenius equation.
Give your answer to an appropriate number of significant figures and include units.

$$\ln k = -\frac{E_a}{RT} + \text{constant}$$

(4)

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(Total for question = 13 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Two chemicals, **E** and **F**, react to form products **G** and **H**.

$$\mathbf{E} + 2\mathbf{F} \rightarrow \mathbf{G} + \mathbf{H}$$

The mechanism for the reaction occurs in two steps via the formation of an intermediate **J**.

$$\mathbf{E} + \mathbf{F} \rightarrow \mathbf{J} \quad \text{slow}$$

$$\mathbf{J} + \mathbf{F} \rightarrow \mathbf{G} + \mathbf{H} \quad \text{fast}$$

What is the rate equation for the reaction?

- ☐ **A** rate = k[**E**][**F**]
- ☐ **B** rate = k[**E**][**F**]^2
- ☐ **C** rate = k[**F**][**J**]
- ☐ **D** rate = k[**E**][**F**]^2[**J**]

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(Total for question = 1 mark)

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Q32.

What can be deduced from the position of the activation energy on a Maxwell-Boltzmann distribution curve?

(1)

- ☒ **A** number of particles in the rate determining step
- ☒ **B** number of successful collisions per second
- ☒ **C** order of reaction
- ☒ **D** proportion of particles with sufficient energy to react

(Total for question = 1 mark)

Q33.

The halogenoalkane, 2-bromobutane, can be hydrolysed using aqueous sodium hydroxide.

Which technique can be used to follow the progress of this reaction?

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(1)

- ☒ **A** colorimetry
- ☒ **B** measurement of gas volume change
- ☒ **C** measurement of mass change
- ☒ **D** titration of quenched samples

(Total for question = 1 mark)

Q34.

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Iodine reacts with propanone under acid conditions. The reaction is first order with respect to propanone, first order with respect to hydrogen ions and zero order with respect to iodine.

What are the units of the rate constant?

(1)

- ☒ **A** $dm^3\ mol^{-1}\ s^{-1}$
- ☒ **B** $mol\ dm^{-3}\ s^{-1}$
- ☒ **C** s^{-1}
- ☒ **D** $mol^{-2}\ dm^6\ s^{-1}$

(Total for question = 1 mark)

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Q35.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Bromate(V) ions, BrO_3^- , react with bromide ions, Br^-, in aqueous acid.

$$BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$$

The rate equation for the reaction is shown.

$$\text{rate} = k[BrO_3^-][Br^-][H^+]^2$$

(a) Which **continuous** monitoring method could be used to obtain kinetics data for this reaction?

(1)

- ☒ **A** colorimetry
- ☒ **B** mass change
- ☒ **C** titration with sodium thiosulfate
- ☒ **D** volume of gas evolved

(b) What are the units of the rate constant, *k*, for this reaction?

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(1)

- ☒ **A** $mol\ dm^{-3}\ s^{-1}$
- ☒ **B** $dm^3\ mol^{-1}\ s^{-1}$
- ☒ **C** $dm^6\ mol^{-2}\ s^{-1}$
- ☒ **D** $dm^9\ mol^{-3}\ s^{-1}$

(c) The concentrations of **all** reactants are halved.

By what factor does the rate of reaction change?

(1)

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☐ **A** $\frac{1}{2}$

☐ **B** $\frac{1}{4}$

☐ **C** $\frac{1}{8}$

☐ **D** $\frac{1}{16}$

(d) The first four steps in the reaction mechanism are shown.

Step 1 $BrO_3^- + H^+ \rightarrow BrO_3H$

Step 2 $BrO_3H + H^+ \rightarrow BrO_3H_2^+$

Step 3 $BrO_3H_2^+ + Br^- \rightarrow Br_2O_2 + H_2O$

Step 4 $Br_2O_2 + Br^- \rightarrow BrO_2^- + Br_2$

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Which is the rate-determining step?

(1)

☐ **A** Step **1**
☐ **B** Step **2**
☐ **C** Step **3**
☐ **D** Step **4**

(Total for question = 4 marks)

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Nitrogen(V) oxide, N_2O_5, decomposes in a first order reaction.

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At 45 °C, the half-life for this reaction is 1400 s.

In an experiment, the initial concentration of nitrogen(V) oxide is 1.0 mol dm^{-3}.

What is the concentration, in mol dm^{-3}, of nitrogen(V) oxide after 4200 s?

☐ **A** 0.875
☐ **B** 0.500
☐ **C** 0.250
☐ **D** 0.125

(Total for question = 1 mark)

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Q37.

A group of students investigated the kinetics of a 'clock' reaction.

The reaction investigated was that between hydrogen peroxide and iodide ions in the presence of acid.

Reaction 1 $H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(l) + I_2(aq)$

In this 'clock' reaction, a fixed volume of aqueous sodium thiosulfate, $Na_2S_2O_3$, and a small amount of starch were added to the reaction mixture.

The added thiosulfate ions react with the iodine produced in **Reaction 1**.

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Reaction 2 $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

When all the thiosulfate ions have reacted, the presence of iodine is detected by the formation of a starchâ€‘iodine complex. The students recorded the time taken for this complex to form.

(a) (i) State the final colour of the mixture containing the starch-iodine complex.

(1)

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(ii) Under appropriate conditions, the reciprocal of time can be used as an approximate measure of the initial rate of the reaction.

Explain why the concentration of the sodium thiosulfate must be low compared with the initial concentrations of the other reagents.

(2)

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(b) Four reaction mixtures, with different initial concentrations of hydrogen peroxide, hydrogen ions and iodide ions, were prepared.

Each mixture had the same volume and contained the same amount of sodium thiosulfate and starch.

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Mixture	$[H_2O_2]$ / mol dm^{-3}	$[H^+]$ / mol dm^{-3}	$[I^-]$ / mol dm^{-3}	Time / s	1 ÷ time / s^{-1}
1	5.4×10^{-2}	1.7×10^{-5}	8.2×10^{-3}	195	5.13×10^{-3}
2	2.7×10^{-2}	1.7×10^{-5}	8.2×10^{-3}	391	2.56×10^{-3}
3	5.4×10^{-2}	1.7×10^{-5}	1.6×10^{-2}	97	1.03×10^{-2}
4	5.4×10^{-2}	1.7×10^{-4}	8.2×10^{-3}	204	4.90×10^{-3}

(i) Use the results in the table to deduce the order of **Reaction 1** with respect to hydrogen peroxide, hydrogen ions and iodide ions.
Justify each answer by referring to relevant data from the table.

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(3)

Hydrogen peroxide .....

.....

Hydrogen ions .....

.....

Iodide ions .....

.....

(ii) Write the overall rate equation for **Reaction 1** using your answers to (b)(i).

(1)

(iii) All four mixtures contained 8.50×10^{-5} mol of sodium thiosulfate. Calculate the amount of iodine that had reacted with the sodium thiosulfate when the colour changed in **Reaction 2**.

(1)

iv) Calculate the rate of reaction, in mol dm^{-3} s^{-1}, with respect to **hydrogen peroxide** using the answer from (b)(iii), the stoichiometry of **Reaction 1** and data from Mixture 1.
The total volume of **each** Mixture was 0.050 dm^3.

(2)

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(v) Calculate a value for the rate constant of **Reaction 1** using data from Mixture 1 and your answers to (b)(ii) and (b)(iv).
Include the units of the rate constant.

(2)

(c) The activation energy for **Reaction 1** may be found by repeating the experiment at different

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temperatures.

One of the students misread the thermometer in their experiment.

ln rate	T / K	$1 \div T$ / K^{-1}
–1.8	333	0.00300
–2.5	323	0.00310
–3.6	308	0.00325
–4.0	307	0.00326
–4.7	291.5	0.00343
–6.0	278	0.00360

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The activation energy, E_a, for a reaction may be found by plotting a graph of ln rate against $1/T$.

Each student carried out an experiment at a different temperature.

The gradient of the resulting line of best fit can be used in the Arrhenius equation to determine a value for E_a, in kJ mol^{-1}.

(i) Determine the value for E_a for **Reaction 1** by plotting a graph using the axes provided.
You should take into account the error made by one of the students.

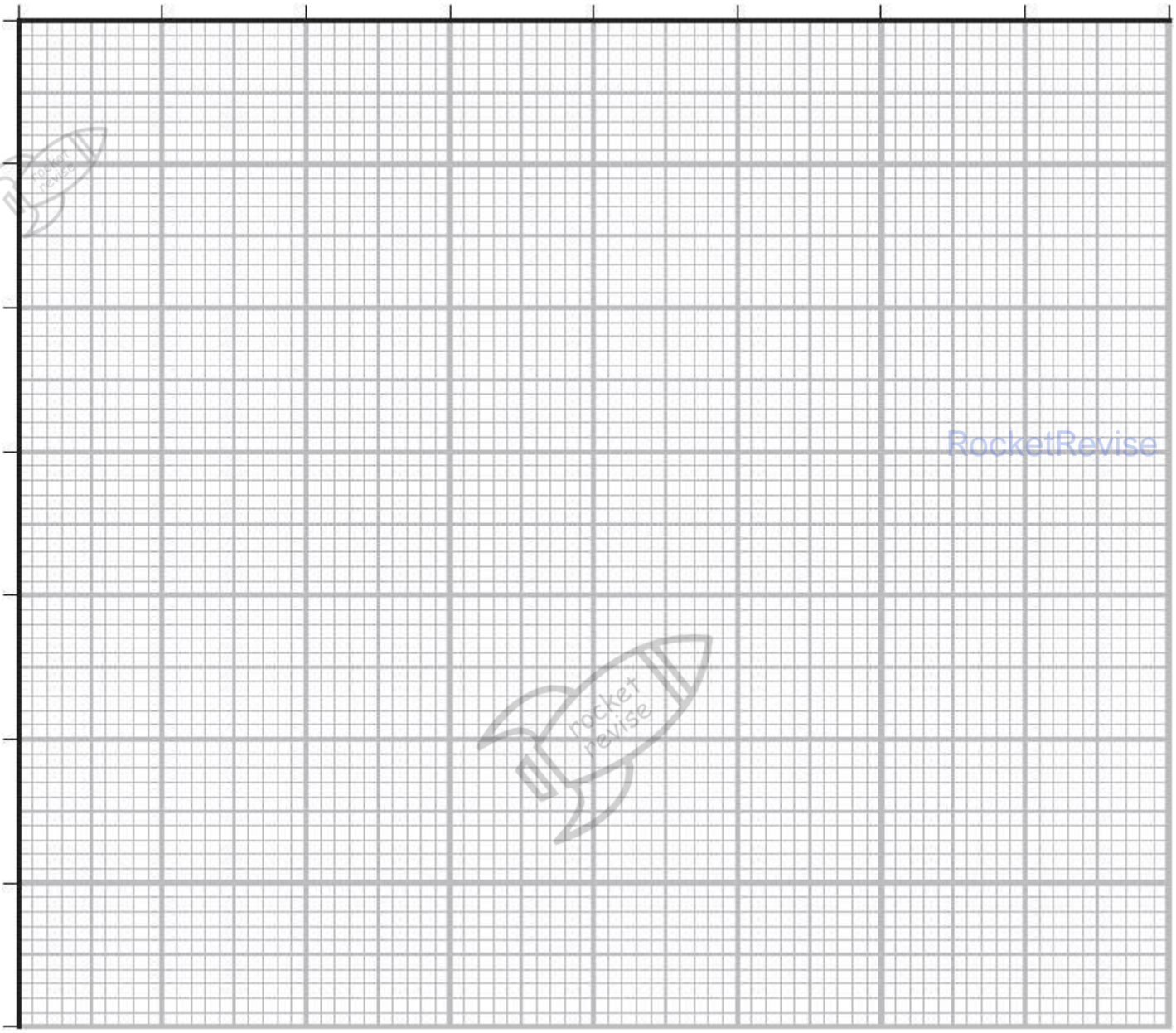
$$\ln \text{ rate} = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant} \qquad R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$$

(5)

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E_a = ... kJ mol^{-1}

(ii) The students all used thermometers capable of reading to the same precision.
Use your graph to deduce the temperature that the student who made the error should have read on their thermometer.

(2)

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(Total for question = 19 marks)

Q38.

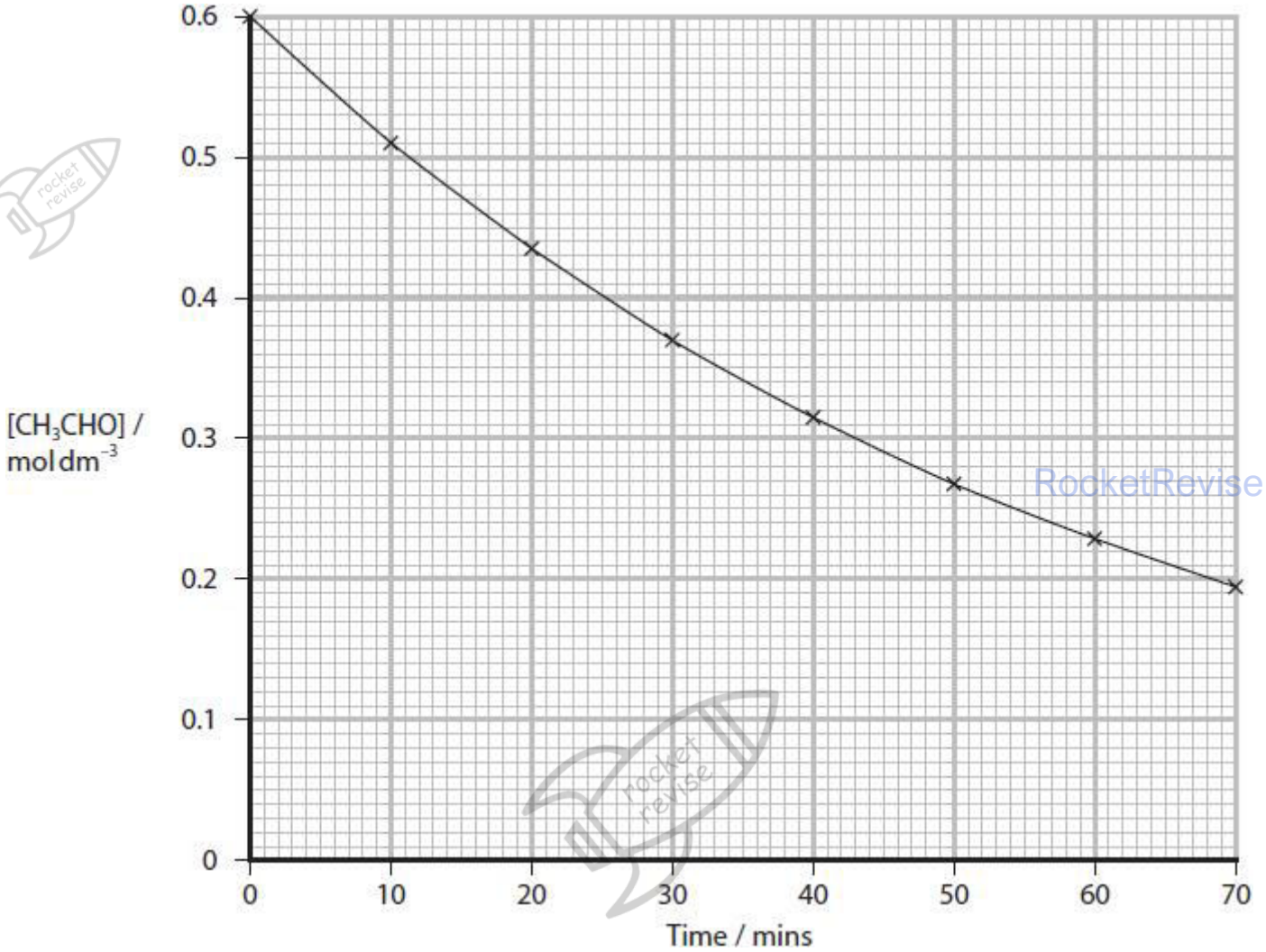
This question is about the kinetics of the gas phase decomposition of CH_3CHO.

$$CH_3CHO \rightarrow CH_4 + CO$$

(a) A concentration-time graph for this reaction at 1000 K is shown.

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(i) Calculate the initial rate of reaction, in **mol dm^{-3} s^{-1}**, at 1000 K.
You must show your working on the graph.

(3)

(ii) Deduce the rate equation for this reaction at 1000 K, by determining two half-lives.
You must show your working on the graph.

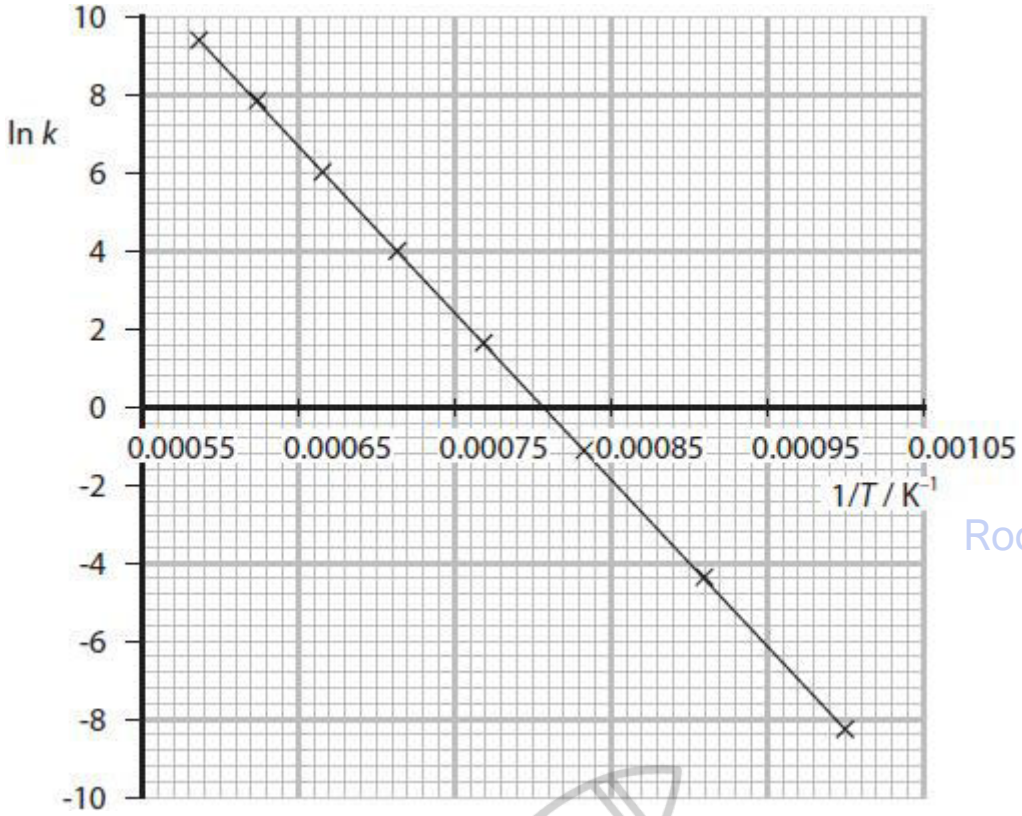
(2)

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(b) The rate constant, *k*, for the decomposition of CH_3CHO was determined between 1000 K and 1700 K.
The results are plotted on the graph of ln *k* against 1 / *T*.

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Determine the activation energy, E_a, for this reaction.

Give your answer to **three** significant figures and include units.

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \text{constant} \qquad R = 8.31\ \text{J mol}^{-1}\ \text{K}^{-1}$$

(4)

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(Total for question = 9 marks)

Q39.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The rate equation for the hydrolysis of a halogenoalkane, RX, in aqueous alkali is shown.

rate = k[RX]

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Which of these statements is **true**?

- ☒ **A** the mechanism involves a carbocation intermediate
- ☒ **B** the reaction has an S_N2 mechanism
- ☒ **C** the halogenoalkane could be 1-bromobutane
- ☒ **D** the main nucleophile in this reaction is H_2O

(Total for question = 1 mark)

Q40.

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This question is about reaction kinetics.

*(a) Halogenoalkanes are hydrolysed by aqueous alkali to form alcohols.

$$RX + OH^- \rightarrow ROH + X^-$$

The rate equations for these reactions with two halogenoalkanes are shown.

1-bromobutane rate = $k[CH_3CH_2CH_2CH_2Br][OH^-]$

2-bromo-2-methylpropane rate = $k[(CH_3)_3CBr]$

Explain how these rate equations can be used to deduce the different reaction mechanisms of the reaction of hydroxide ions with these halogenoalkanes.

(6)

(b) An initial-rates method may be used to investigate the kinetics of the reaction between hydrogen peroxide and iodide ions in acid solution. The equation for this reaction is shown.

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

The reaction mixture contains a small amount of thiosulfate ions which reduce the iodine back to iodide ions. The equation for this reduction is shown.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

As soon as the thiosulfate is used up, the iodine that is further produced reacts with the starch present and results in a colour change. The time taken for this colour change is noted and the reciprocal of this time is used as a measure of reaction rate.
The results from a series of experiments where the volume of the hydrogen peroxide was changed are given in the table.

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Reaction rate / s^{-1}	0.9×10^{-2}	1.8×10^{-2}	2.5×10^{-2}	3.7×10^{-2}	4.6×10^{-2}
Volume of H_2O_2 / cm^3	10	20	30	40	50

(i) Give the colour **change** after all the thiosulfate is used up.

(2)

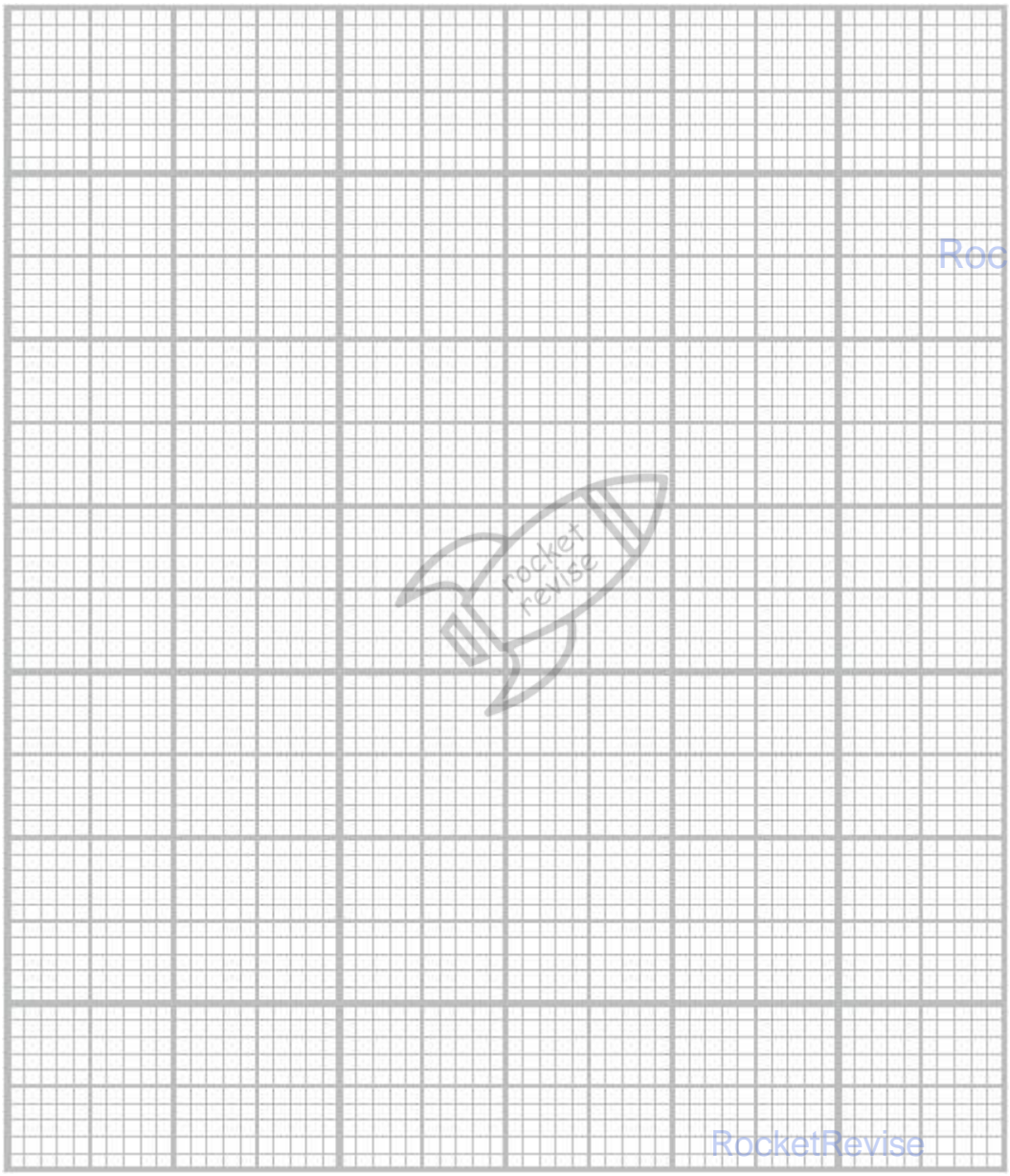
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(ii) Plot a graph of reaction rate against volume of hydrogen peroxide.

(3)

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(iii) State and justify the reaction order with respect to hydrogen peroxide by using your graph.

(1)

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(c) The continuous monitoring method was used to investigate the kinetics of the reaction between magnesium and sulfuric acid at different temperatures.

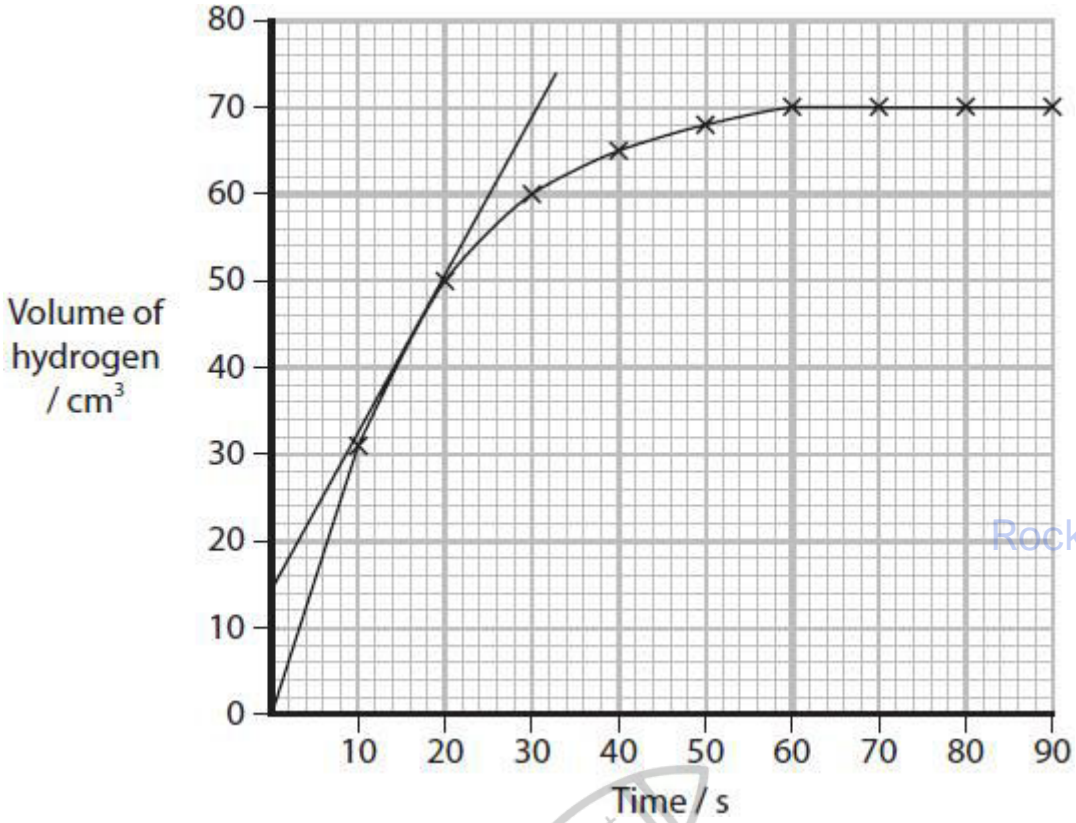
The equation for the reaction is shown.

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

(i) The graph shows the results of an experiment at 80°C.

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Calculate the reaction rate at 15 seconds using the tangent shown.
Give your answer to two significant figures and include units, if any.

(2)

(ii) The temperature of the reaction was changed to 75°C.
Draw a line on the graph to show the possible results if this was the only change. Justify any similarities and differences in the lines.

(2)

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(Total for question = 16 marks)

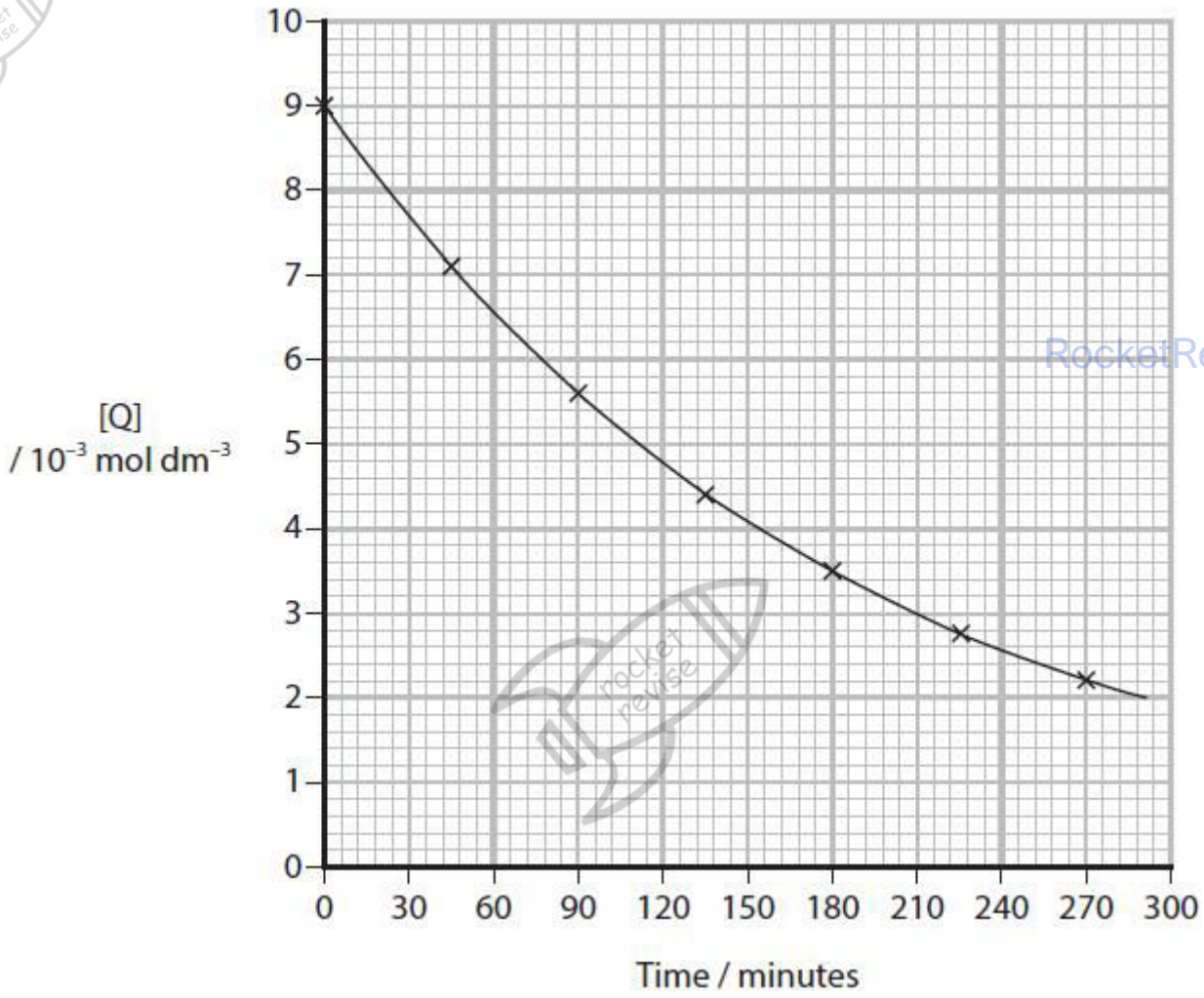
Q41.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A graph of the concentration of Q during decomposition is shown.



What is the half-life for this decomposition?

- ☒ **A** 93 minutes
- ☒ **B** 132 minutes
- ☒ **C** 146 minutes
- ☒ **D** 291 minutes

(Total for question = 1 mark)

Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

In an experiment to determine the rate of reaction between bromine and methanoic acid, which method would **not** be suitable?

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$$Br_2(aq) + HCOOH(aq) \rightarrow 2H^+(aq) + 2Br^-(aq) + CO_2(g)$$

- ☒ **A** quenching and titrating with an acid
- ☒ **B** measuring the loss in mass
- ☒ **C** colorimetry
- ☒ **D** measuring the volume of gas produced

(Total for question = 1 mark)

Q43.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

A chloroalkane, RCl, undergoes hydrolysis according to the equation shown.

$$RCl + OH^- \rightarrow ROH + Cl^-$$

The rate equation is rate = k[RCl]

Which chloroalkane is most likely to be RCl?

- ☒ **A** CH_3CH_2Cl
- ☒ **B** $(CH_3)_3CCl$
- ☒ **C** $CH_3CHClCH_3$
- ☒ **D** $(CH_3)_3CCH_2Cl$

(Total for question = 1 mark)

Q44.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

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The reaction shown is second order with respect to NO and first order with respect to O_2.

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

If the concentration of NO is doubled and the concentration of O_2 is tripled, by what factor will the rate increase?

- ☒ **A** 6
- ☒ **B** 12
- ☒ **C** 18
- ☒ **D** 27

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(Total for question = 1 mark)

Q45.

This question is about the kinetics of the catalytic decomposition of hydrogen peroxide, H_2O_2, using manganese(IV) oxide as a catalyst.

The equation for the decomposition is shown.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

In an experiment, $10cm^3$ of hydrogen peroxide solution was added to $150cm^3$ of distilled water. A small amount of manganese(IV) oxide granules was added and the clock started.

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Every 5 minutes, $10cm^3$ samples of the reaction mixture were withdrawn using a pipette and transferred to a conical flask.

The samples were then acidified with dilute sulfuric acid and titrated with potassium manganate(VII) solution. The equation for the titration is shown.

$$2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$$

The results of the experiment are given in the table.

Time / min	0	5	10	15	20	25	30
Volume of manganate(VII) / cm^3	32.5	24.8	18.9	13.9	10.5	8.1	6.5

(a) (i) The reaction in the sample stops immediately on removal from the reaction mixture.

Suggest why this occurs.

(1)

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(ii) State why it is not necessary to calculate the concentration of hydrogen peroxide solution when trying to deduce the order of reaction with respect to hydrogen peroxide.

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(1)

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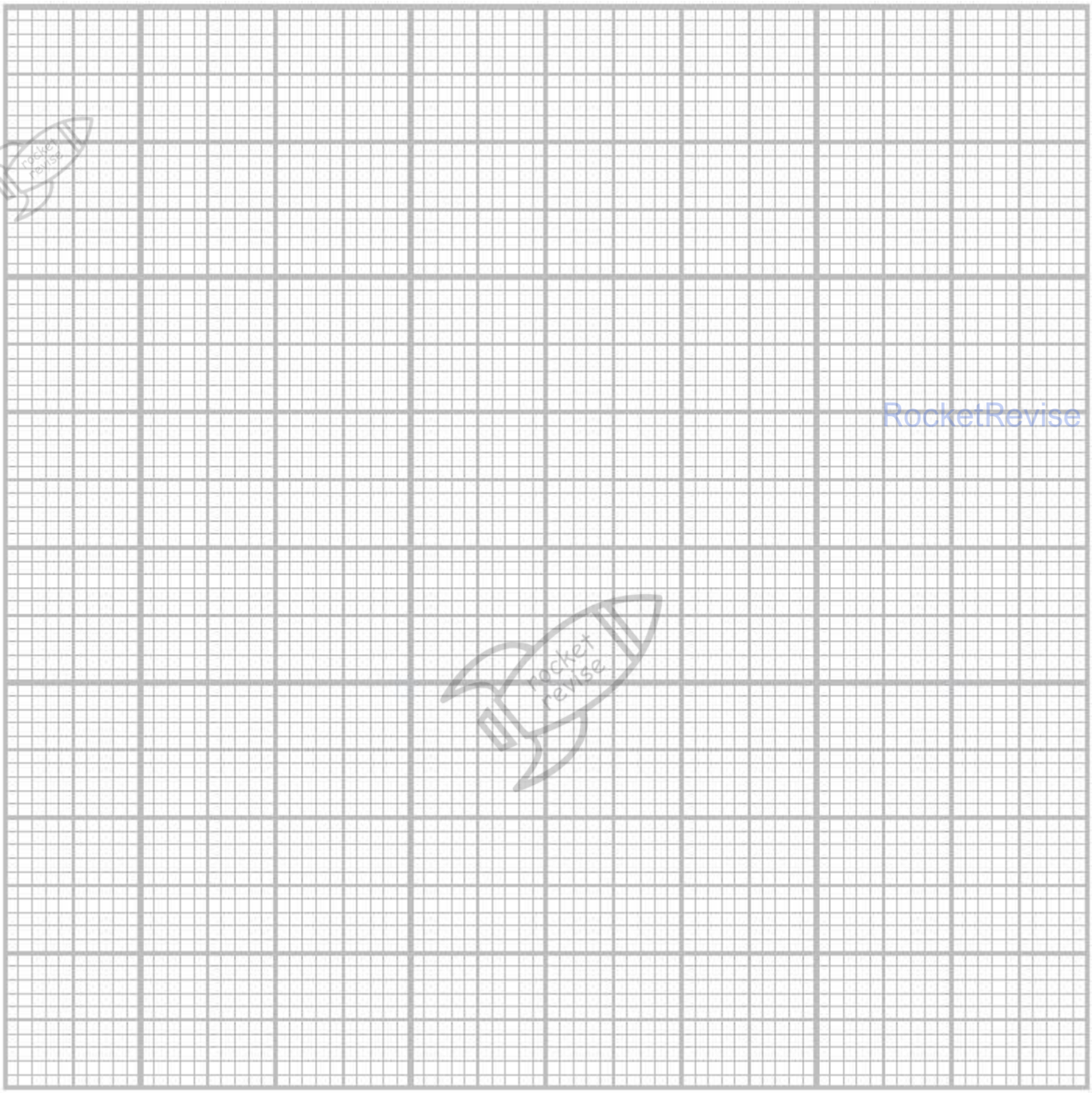
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(iii) Plot a graph of the volume of potassium manganate(VII) solution against time.

(3)

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(iv) Use your graph to show that the reaction is first order with respect to hydrogen peroxide. You must show your working on the graph.

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(3)

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(b) Two possible mechanisms have been suggested by students carrying out this reaction.

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Mechanism 1

Step 1 $2H_2O_2(aq) \rightarrow [H_4O_4](aq)$ slow
transition state

Step 2 $[H_4O_4](aq) \rightarrow 2H_2O(l) + O_2(g)$ fast

Mechanism 2

Step 1 $H_2O_2 \rightarrow 2OH$ slow

Step 2 $H_2O_2 + OH \rightarrow H_2O + HO_2$ fast

Step 3 $HO_2 + OH \rightarrow H_2O + O_2$ fast

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Explain whether or not these mechanisms are consistent with the reaction being first order.

(2)

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(Total for question = 10 marks)

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